

2016.0 RANGE ROVER (LG), 412-05

REFRIGERATION

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the Refrigeration system, refer to the relevant Description and Operation section in the workshop manual.

REFER to: Refrigeration (412-05, Description and Operation).

INSPECTION AND VERIFICATION

CAUTIONS:

- Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.
- When probing connectors to take measurements in the course of the pinpoint tests, use the adaptor kit, part number 3548-1358-00

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the warranty policy and procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places and with a current calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account
- If diagnostic trouble codes are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Bottle chiller door	▪ Fuses ▪ Relays ▪ Loose or corroded connectors(s)

- Bottle chiller
- Rear Seat Entertainment Control Module (RSECM)
- Bottle chiller switch

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step
4. If the cause is not visually evident, check for Diagnostic Trouble Codes (DTCs) and refer to the DTC index

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Bottle chiller inoperative (switch status lamp on or off)	▪ Bottle chiller power or ground circuit open circuit, high resistance ▪ Bottle chiller switch circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Bottle chiller relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Bottle chiller internal failure	NOTE: Fuse 21P ▪ Refer to the electrical circuit diagrams and check the bottle chiller power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the bottle chiller switch circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the bottle chiller relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Install a new bottle chiller
Bottle chiller inoperative (switch status lamp flashing)	▪ Rear seat entertainment control module voltage greater than 18.0V ▪ Rear seat	

	entertainment control module voltage less than 11.0V	NOTE: Bottle chiller function will be restored when the voltage returns to between 11.5V and 15.5V - Refer to the relevant section of the workshop manual and test the battery and charging system - Refer to the electrical circuit diagrams and check the rear seat entertainment control module power and ground circuits for open circuit, high resistance. Refer to the relevant section of the workshop manual and test the battery and charging system
Bottle chiller lamps inoperative	- Bottle chiller lamps circuit short circuit to ground, short circuit to power, open circuit, high resistance - Rear seat entertainment control module internal failure	NOTE: The bottle chiller lamps operate only when the bottle chiller door is open - Refer to the electrical circuit diagrams and check the bottle chiller lamps circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Install a new rear seat entertainment control module

REAR SEAT ENTERTAINMENT CONTROL MODULE TEST MODE

The rear seat entertainment control module does not set DTCs. Instead, there is a test mode which is used to indicate some of the rear seat entertainment control module signals.

NOTES:

- Test mode can only be entered within 15 seconds of ignition on.
- Test mode will be exited if there are no further switch operations within 30s, or the ignition is set to off.

1. Set the ignition to on

2. Operate either table switch five times while operating either iPad™ audio select switch (this must be completed within 2 seconds)

Check the operation of the relevant component with reference to the table below

COMPONENT	TEST MODE SIGNAL
Ambience lighting	On
C pillar interior lamps	On
Bottle chiller relay	Cycled every 2 seconds
Bottle chiller switch status lamp	On when the switch is operated
Left iPad™ audio select switch status lamp	On or off to indicate powered table home switch status
Right iPad™ audio select switch status lamp	Flashes when powered table is raised or lowered

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.